Panel Applications and Midterm 2 Review

09 Nov 2026 | Chapter 10

ECO 351 – Introduction to Econometrics

Shi Qiu – Fall 2026

Quiz 4

1. Open Quiz 4 in Blackboard and answer all six questions.
2. The entire quiz is worth four points.
3. Follow the Blackboard directions and submit when instructed.

Stata: the same sample, three comparisons

```
. estimates table pooled statefe twoway, b(%9.3f) se(%9.3f) stats(N)
-------------------------------------------------
    Variable |  pooled      statefe     twoway
-------------+-----------------------------------
     beertax |     0.365      -0.656      -0.640
             |     0.120       0.292       0.357
-------------+-----------------------------------
           N |       336         336         336
-------------------------------------------------
                                     Legend: b/se
```

Selected raw rows. `pooled`: pooled OLS. `statefe`: state effects. `twoway`: state and year effects. The second row for each coefficient reports its state-clustered SE.

Exercise: explaining the change of sign

A report states: "The pooled tax coefficient is positive and the fixed-effects coefficient is negative. Clustering must have reversed the sign."

1. Explain the mistake.
2. Name the variation behind the pooled and state-effects estimates.
3. Give an example of a persistent omitted state characteristic that could contribute to the difference.

Debrief: specification and uncertainty

Changing the standard-error calculation leaves the pooled OLS coefficient unchanged.

State fixed effects change the comparison by removing each state's persistent baseline. Year fixed effects additionally account for common shifts across years.

Remaining state-specific changes can still confound the estimate. A sign reversal supplies a reason to investigate the specification, not a causal conclusion by itself.

Stata: a ten-cent policy comparison

```
. lincom 0.10*beertax
 ( 1)  .1*beertax = 0
------------------------------------------------------------------------------
    fatality | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   -.063998   .0357078    -1.79   0.080    -.1358329    .0078369
------------------------------------------------------------------------------
```

The interval crosses zero. Report the fitted rate difference and its uncertainty, rather than a definite number of lives saved.

Exercise: a threat that fixed effects leave behind

A state raises its beer tax and introduces extra road-safety enforcement in the same year. Both could reduce fatalities.

Explain why state effects and common year effects may leave this concern unresolved. What additional information would you want before making a causal claim?

Fixed effects, random effects, and a Hausman comparison

Both models allow a state-specific baseline. **Random effects** additionally assumes that baseline is unrelated to the included regressors. **Fixed effects** permits that correlation.

A classic Hausman comparison tests whether the FE and RE slope differences are systematic under its maintained assumptions.

- Null: RE is consistent under the maintained model.
- Rejection supports FE relative to RE.
- Non-rejection does not prove the RE assumption or a causal interpretation.

Stata: a classic Hausman comparison

```
. hausman fixed random, sigmamore
Test of H0: Difference in coefficients not systematic
    chi2(1) = 16.86
Prob > chi2 = 0.0000
```

The do-file estimates compatible FE and RE models without clustered VCEs for this classic test, then re-estimates FE with state-clustered standard errors.

Review: the estimand and the regression

A school regression includes STR, income, and the English-learner share. A researcher wants the effect of reducing STR by two.

1. What must be held fixed in the fitted comparison?
2. In an additive model with STR coefficient -0.64, what is the fitted change?
3. How could a targeted funding policy make the fitted association differ from the causal effect?

The coefficient in this exercise is constructed.

Debrief: a fitted change and its causal interpretation

$$\Delta\widehat{Score} = (-0.64)(-2) = 1.28 \text{ points.}$$

The calculation holds the included controls fixed. A causal interpretation additionally needs the remaining score determinants to be unrelated to the relevant STR variation.

If unmeasured targeted funding changes both STR and outcomes, the arithmetic remains correct for the fitted model while the causal interpretation can fail.

Review: a nonlinear fitted comparison

In a constructed model, income I is in thousands of dollars and

$$\widehat{Score} = 620 + 3I - 0.04I^2 - 0.5STR.$$

Compare income 20 with 25 while holding STR fixed.

1. Write the coefficient weights before calculating the change.
2. Explain why 3×5 alone is incomplete.

Debrief: both income terms change

$$\Delta\widehat{Score} = 3(25 - 20) - 0.04(25^2 - 20^2) = 6.$$

The intercept and fixed STR contribution cancel. The square changes by 225, rather than $(25 - 20)^2 = 25$.

The six-point result is a fitted difference. The equation alone does not establish a causal income effect.

Review: a joint restriction

A constructed wage model includes education, experience, and experience squared. You want to test whether the two experience terms jointly contribute to the population regression.

1. State the joint null and the number of restrictions.
2. A joint test gives $p = 0.03$. What is the 5% decision?
3. Does rejecting that null show the regression is causal?

Debrief and project reminder

The joint null sets both experience coefficients to zero, so there are two restrictions. With $p = 0.03$, reject at 5%. Identification remains a separate issue.

Term project due today, 9 November: submit the complete package through iLearn under the existing instructions.

Next teaching meeting, after Midterm 2: regression when the outcome is zero or one.